

Experience

JUMP TRADING starting Jun 2026
Vendor Data Group Software Engineer summer intern.

NTT DATA (UCL IXN) Oct 2025 – Mar 2026
Carbon-aware AI workload scheduling. Optimal resource allocation webservice implemented in C++23 with policy-based parallelism, x86 SIMD, and coroutines. [Repo](#)

UCL Sep 2025 – Mar 2026
Teaching Assistant and my course's Programming Tutor.

Extracurricular

UCL COMPETITIVE PROGRAMMING SOCIETY Nov 2024 – Present
Founding & committee member and speaker. We host topical lectures, contests, and training sessions with 130 members.

JESS AI CLUB Jan 2022 – May 2022
Founder and host of the first purely student-ran club. Its popularity allowed me to introduce NLP theory to 20+ peers.

ACADEMIC RESEARCH May 2023 – Oct 2023
Co-authored a [16-page report](#) on water usage in the UAE. Presented at the [Roots & Shoots Expo Awards Ceremony](#).

Competitions

COMPETITIVE PROGRAMMING Jan 2025 – Present
[11th](#)/310 (1st in London) at the 2025 [UKIEPC](#)
[53rd](#)/141 at the 2025 [ICPC NWERC](#)
2nd at the 2025 [Recraft Mini-Cup](#)
Codeforces: 16th in UK (Mar 2026). Leetcode: top 7.61%.

MATHEMATICAL MODELING Nov 2022; Feb 2023
Awarded *Meritorious* in [HiMCM](#) and *successful in triage* in [M3C](#). Led and trained my own team. Performed time-series analysis, non-linear and periodic regression, and dynamics modeling using SciPy and Mathematica.

Projects

[gws2](#) ([akioweh.com](#)) Oct 2024 – Present
A compact server on continuous deployment prototyped with modern technologies like 0-RTT h3 and reverse proxying.

Team-based Software Engineering Oct 2025 – Dec 2025
An academic management system using Spring Boot & JPA with React developed concurrently as a team of three.

[OiqBot](#) Oct 2021 – Present
Real-time network app with a cooperative concurrency model and modular, hot-loadable plugins. Engineered a non-blocking (de)initialization procedure for quick restarts.

[sheepcalc](#) Jul 2025 – Jul 2025
Pure λ -calculus interpreter in Haskell using ADS and Parsec.

[AEQ](#) May 2024 – Mar 2025
Audio EQ manipulation tool. Supports parametric cascaded-biquad IIR and phase-controlled convolution FIR.

Education

UCL 2024 – 2027
BSc Computer Science Y1: 83.38%

JESS Dubai 2020 – 2024
GCSEs (11), IB Diploma

Technical Skills

ALGORITHMS & DATA-STRUCTURES
Competition code; Technical analysis

APPLICATION PROGRAMMING
Concurrency; RDBMS; REST API;
ASGI; Template programming; Scalable real-time applications

LANGUAGES
Python (5yr): FastAPI, SciPy, aiohttp
C++ (3yr): c++23, CMake, drogon;
coroutines, concepts, value semantics
Java (3yr): Maven; JSP & JSTL, Spring
Haskell (1yr); Lua (1yr)

NETWORKING
WAN Architecture, Anycast; DNS,
email security: DNSSEC, DMARC;
NAT, Subnets; LAN administration

DEVOPS & SYSADMIN
Cloud infrastructure / VPS, Terraform;
Enterprise & SE Linux; CI/CD

MICROCONTROLLER & EEE
ESP-IDF, PlatformIO; UART, I2C;
Control Dynamics

DATA SCIENCE
Computational modeling: Time-series,
Regression, Monte Carlo, Dynamics;
Visualization; Validation & Forecasting

DEVELOPMENT TOOLS
Git, CMake; LSP, DAP; MSYS2,
Archlinux; NeoVim, ~~JB~~ IDEs, SSH

TYPESETTING
Typst; L^AT_EX, T_ikZ, BibT_EX

Misc

LIVE AUDIO & VIDEO
Event audio-visual control & setup;
Live mixing and audio engineering

NATURAL LANGUAGES English, Mandarin